



MANIPAL
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MDS5105/ MSBA5105 Segment 01

INTRODUCTION TO DATA AND INFORMATION

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Introduction

Data resides at various sources. To extract information from data, it is required to pool together all the data from multiple sources and put them in a common location. Therefore, all the data from different sources needs to be brought into the same repository and then the data undergoes processing. In data processing, data needs to be transformed for one or more distinct reasons. This is required because the data is taken from various sources and the format of data representation may be different. There are several other reasons due to which data needs to undergo processing.

Once the data is processed, the streamlined pattern of the data will provide the information. Hence, the information is the set of data that will be used for further knowledge discovery.



Learning Objectives

At the end of this topic, you will be able to:

- Explain the importance of data
- Identify the need for data processing
- Outline the various data sources
- Summarise the ecosystem required for data processing and decision making

1. Importance and Need for Data

Terms such as artificial intelligence, data analysis, and data mining have become part of common parlance today; "data" is often used and heard. Data plays a significant role in all aspects; hence it is worth knowing the importance of data so that working on the same can be better transferred into valuable insights.

The importance of data is as follows:

1. Data improves people's lives. The numerous gadgets and applications which we are using today are data driven. There are smartwatches and smartphones that keep track of our daily vitals, such as, heart rate, calories' count, and even the quantity of water we drink. Besides these, there are several other tracking applications which help improve people's lives.
2. Data is very useful in making decisions for any organisation.
3. Organisations today consider data as an asset and use it for planning their strategies.
4. Based on the strategies and decisions made, data is used in predicting the outcomes of strategies and decisions that are made by the organisations.
5. Data helps monitor real-time data and track those which are monitored.
6. Data is useful in accessing resources where it will help in the reusability of these data which are used from the various sources for various applications.

2. Data, Information, Knowledge, and Wisdom

Streamlined data provides information, and this information, which is extracted from the data, leads to the discovery of knowledge. Knowledge refers to being informed about a subject. It helps us in gaining wisdom; and wisdom is the ability to make a judgment.

When there is discovery of knowledge, there arises some critical questions such as, where to implement that knowledge, why to implement, and why not to implement. These

questions should be resolved with wisdom which, in turn, comes with experience and knowledge.

Table 1: Various Types of Data

Data	5	Colleges	Cities	1980	Mangoes	Born	Top	Quality	Mr. XYZ
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For example, in table 1, we have a set of data, i.e., 5, Colleges, Cities, 1980, Mangoes, Born, Top, Quality, and Mr. XYZ. These are some random numbers, words, and some characters. This is just data; hence, it is not going to provide any information at this point. Only a streamlined pattern of data can provide information. so, we can try to find some pattern in this data, say, 5 top colleges in the cities; Mr. XYZ was born in the year 1980, etc. Now, the set of data is giving some information. This is how streamlining of data works towards providing information.

We gain knowledge from this information, i.e., the streamlined pattern of the data or the well-organised body of information. And, with this body of knowledge and experience, we gain wisdom.

Let us look at another example:

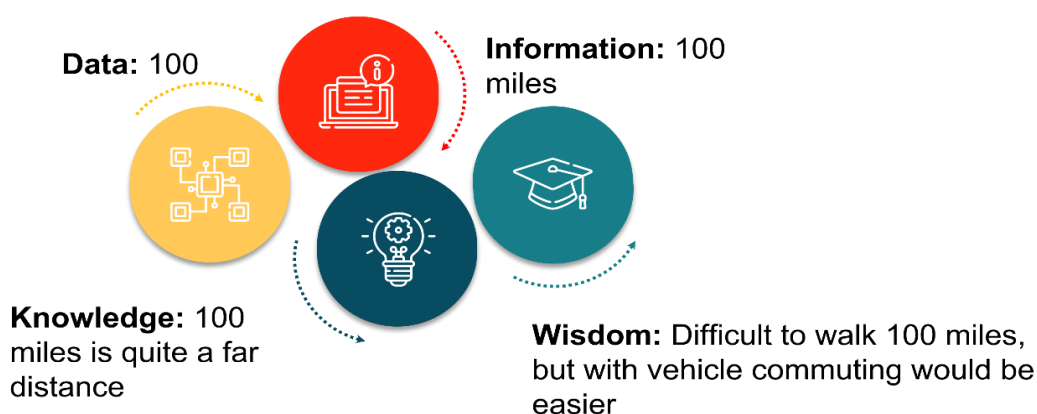


Figure 1: Gaining Wisdom from Data

In the above example, the data is the number 100. Let us find some pattern from the given number. Suppose if we add miles to it, the number can be patterned as 100 miles. Now, we have the information that we are discussing the distance. So, 100 miles is the information, and the knowledge we gain is that 100 miles is quite a long distance. Thus, the person with the wisdom can say that 100 miles is difficult to walk, but with a vehicle, commuting would be easier—this is called wisdom!

This example shows how wisdom can be gained based on experience and knowledge, how the knowledge can be extracted or discovered from information, how information is patterned, and how data is streamlined to arrive at that information.

3. Data Collection

Let us now find out how to collect data, how to look into the various sources, where to get data from, why do we need data, and what data do we need.

We will start with the data collection techniques. Some of the data collection techniques are:

1. Oral history
2. Online marketing
3. Interviews
4. Questionnaires
5. Observation
6. Documents and records
7. Focus group
8. Social media monitoring

Oral history is one of the ways of collecting data which refers to a well-informed interviewer and a narrator with first-hand knowledge of historically significant events engaging in recorded interviews as a way of conducting historical research with the aim of enhancing a

historical record. Another method is online marketing. In online market research, the process of gathering data is done online.

Interviews with experts also help in collecting the data. In this method, data is to be collected based on the intention of the study or what we intend to do. We choose the appropriate data collection method accordingly.

Then, there are questionnaires; these refer to the questions prepared to collect the data. These questionnaires are distributed among participants in focus groups. Focus group discussions are widely used as a qualitative strategy to gain in-depth understanding on social topics. The technique tries to get information from a carefully chosen sample of people. Subsequently, information is gathered from these people.

Observational studies also provide data. This method refers to collecting data by observing individuals in a certain location or environment on a particular day and time. Data can also be collected from various documents and records. These can be in the form of hard copy or electronic form, different kinds of logs, journals, communications, movies, computer databases, audio, or digital records, or even the actual results of studies.

Besides these, social media monitoring provides valuable data. Social media monitoring refers to monitoring of followers' interaction on a brand's social media accounts. It is a simple method to keep a track on information about the motives and interests of your audience. Although many social media platforms come with statistics, there are other third-party social media sites that provide more thorough, organised insights gleaned from many channels.

4. Data Processing

Once data is collected, all the data is put together in a bucket. And then, that data is cleaned and processed. It undergoes some transformations, such as, filtering, segregation, normalisation, cleaning, etc.

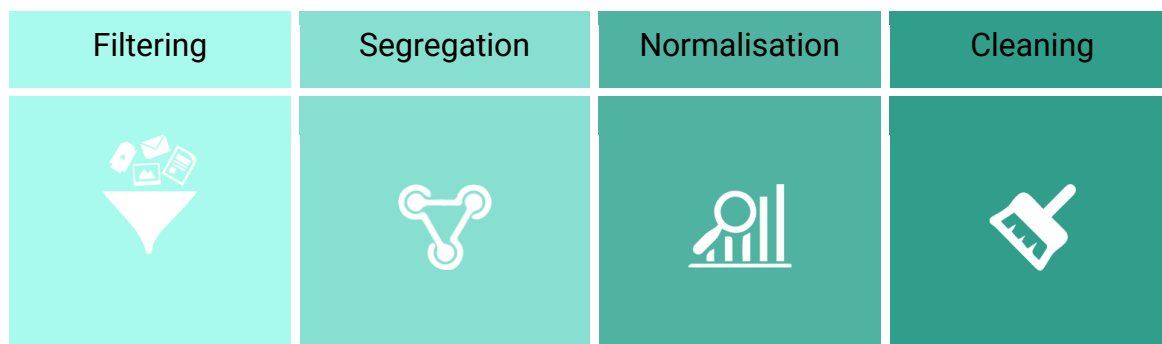


Figure 2: Data Transformation

This process depends on the intention of the study or what kind of analysis or visualisation we want to do in the later stages. All the data that is put into one bucket before cleaning and processing are raw data, and the processed data or transformed data will be put into another bucket for further process.

5. Knowledge Discovery

Identifying the pattern of the processing and discovering is an iterative process. A pattern is streamlined data that is represented in various patterns. Once, those patterns are identified, we discover knowledge.

This is further represented in the form of reports, tables. characterisation rules, etc.

6. Information System

In the present times, we are aware of Database Management System in which our data is put into a database. The term database relates to the keyword 'information system'. The innovation of information system into various domains in the past couple of decades has helped us in populating enormous amount of data that we are using today. This is helping us in terms of analysis and decision making. Therefore, information system plays several roles in handling data or storing data.

6.1 Stakeholders

Information system plays a vital role in populating data, and we need to consider who are the stakeholders of information system. Owners are one of the stakeholders and users are also considered stakeholders because it not always the owners who are the only users of an information system. This could be understood with an example. Suppose we visit a nearby pharmacy. They have an information system where they store all their stocks and selling points and manage all the accounts. In this case, the owners of the pharmacy are not the only users of the information system that they have. The customers are also a part of the information system. The stocks are part of the information system as well. Hence, the owners and the users are different.

Based on the requirements of the users and owners, designers design the required information system, and developers implement and develop the information systems. Then, there are system analysts who monitor and analyse, or track the information systems, and then, we have managers who will be looking after the information system.

7. Data Warehouse

The next important term is data warehouse. We know that data is collected from various sources, and one of them can be a database. The data undergoes further processing, and then the processed data is put into a bucket. The bucket here refers to data warehouse. Data warehouse is designed to enable and support business intelligence activities, especially analytics.

Data warehouse holds all the processed data, i.e., the data which are curated and transformed. This data warehouse is intended to be used to perform queries and analysis and often contains a large amount of historical data. The data warehouse makes data mining or analysis possible, which in turn assists businesses to look for data patterns. So, we can conclude that the data that resides at a data warehouse is the curated data or processed data that is prepared to be analysed.

8. Data Sources

Other data sources apart from data warehouse are:

1. Data lake
2. Data mart

Let us look at the difference between data lakes, data warehouses, and data mart.

Data warehouse holds the data which is curated and is ready to be used for analysis. While, a data lake is just a vast pool of raw data, the purpose of which is not yet defined. On the other hand, in the case of the data which resides in a data warehouse, we know the purpose of the data that we will be using for and the type of analysis or the type of study the data will be used for.

Data warehouse is a repository for structured and filtered data that has already been processed for a specific purpose. Data mart is very much focused when compared to the data warehouse. It is a subset of a data warehouse that contains repositories of summarised data collected for analysis on a specific section of the unit. Within the data warehouse, we can have an “n” number of data marts. Hence, it is a subset of data warehouse which is extremely focused compared to the data which are in data warehouse.

Table 2: Differences Between Data Lakes, Data Warehouse, and Data Mart

	Data Lake	Warehouse	Mart
Scope	Broad	Cleaned	Focused
Pre-processing	Low	Moderate	High
Analysis limitation	Only limited to input sources	Limited by data cleaning choices	Limited to focus topic
Ease of navigation	Hard	Moderate	Easy

While comparing data lake, data warehouse, and data mart based on the four items—scope, pre-processing, analysis limitation, and ease of navigation, the scope of data lake is very broad. The scope of warehouse is clean, and that of data mart is focused.

It can be said that data lake and mart are two extremes. The scope of data lake is broader and the data in mart is very focused, or we can say, it is narrowed down to specific functionalities. The data which resides in data lake undergoes very low pre-processing and the data in warehouse undergoes moderate pre-processing, whereas the data in mart undergoes a high level of pre-processing. Analysis limitations of data residing in data lake are only limited to the input sources, in warehouse, it is limited by data cleaning choices and in data mart, it is limited to the focus of the topic. The navigation of data is very hard in data lake, moderate in data warehouse whereas it is very easy in data mart.

Data warehouse, data lake, and data mart are three various data storages used during data mining, data analysis, and data visualisation.

9. Summary

Here is a quick recap of what we have learnt so far:

- Data from various sources are gathered to process and generate reliable patterns.
- Data gives information, from which knowledge is discovered, and it leads to achieving wisdom.
- A data lake is a vast pool of raw data, the purpose of which is not yet defined.
- A data warehouse is a repository for structured, filtered data that has already been processed for a specific purpose.
- A data mart is a subset of a data warehouse and contains repositories of summarised data collected for analysis on a specific section or unit.

10. Glossary

Information	The streamlined data gives information.
Knowledge	It is being informed about the subject.
Wisdom	It is the ability to make a judgment.
Data Lake	It is a vast pool of raw data, the purpose for which is not yet defined.
Data Warehouse	It is a repository for structured, filtered data that has already been processed for a specific purpose.
Data Mart	It is a subset of a data warehouse and contains repositories of summarised data collected for analysis on a specific section or unit.

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